

Dynamic Network Formation for Robust Decentralized Personalized Learning over Hybrid 5G C-V2X Vehicular Networks

IEEE Publication Technology, Staff, IEEE,

Abstract—Connected vehicles increasingly rely on collaborative intelligence for services such as cooperative perception, risk prediction, and personalized driver assistance. Realizing such intelligence requires learning from distributed, privacy-sensitive, and strongly non-IID data under stringent latency, energy, and bandwidth constraints. Existing federated and edge-assisted methods remain poorly suited to this setting: they rely on centralized orchestration, assume a single communication interface, and offer limited resilience to unreliable peers. We study robust decentralized personalized learning over hybrid 5G C-V2X networks, whose PC5 sidelink and Uu interfaces offer complementary range, latency, and cost profiles for resource-aware collaboration. We formulate collaboration as a dynamic directed network formation game in which each vehicle chooses its collaborators, communication mode, and aggregation weights under local resource and trust constraints. Based on this formulation, we propose Dante, a fully decentralized framework in which every vehicle learns a link-formation policy through an attention-guided actor-critic, aggregates received models under fused attention-trust-robustness weights, and maintains online local trust estimates for its peers. We analyze convergence and per-round complexity, and SUMO-driven experiments over real urban and highway maps show that Dante delivers substantial resource savings and more efficient collaboration while preserving personalized accuracy and Byzantine robustness over representative baselines.

Index Terms—Decentralized federated learning, personalization, vehicular networks, 5G C-V2X, network formation, Byzantine robustness.

I. Introduction

Modern connected vehicles are increasingly equipped with high resolution cameras, LiDAR, radar, telemetry modules, and driver or cabin monitoring systems that continuously observe both the external traffic scene and the internal vehicle context [1], [2]. Consequently, vehicles generate massive volumes of multimodal data, with recent surveys reporting data rates that can reach the terabyte per hour scale in advanced autonomous driving settings [1], [2]. Exploiting such data enables a broad class of learning tasks, including collision risk estimation, trajectory forecasting, cooperative perception, and personalized driver assistance, all of which require models that are both context aware and responsive to local driving conditions [3].

Traditionally, such models have been trained by uploading vehicle data to cloud servers, where a global model is learned over the aggregated corpus. This paradigm is

impractical in current vehicular environments as transferring massive amounts of data is prohibitively expensive both in terms of bandwidth and energy. Furthermore, it conflicts with strict latency constraints and automotive privacy regulations. Federated learning (FL) partially addresses this tension by enabling collaborative training through model exchange rather than raw data sharing [4], [5]. Classical FL, however, remains fundamentally server centric. Its dependence on a central aggregator introduces a communication bottleneck and a single point of failure, while the learned objective is still biased toward a common model that may not adequately serve vehicles operating under distinct traffic regimes, road geometries, sensing conditions, and driver preferences.

To reduce dependence on remote cloud coordination, several vehicular learning systems have shifted aggregation to roadside units, base stations, or edge servers near the road network [6]–[8]. Although this architecture lowers communication latency, it does not remove central orchestration. Vehicles must still maintain infrastructure connectivity, contend for shared uplink resources, and often contribute to regional or global models whose inductive bias remains insufficiently personalized. Mobility further complicates this design because short contact times, frequent handovers, and intermittent coverage can degrade both training efficiency and aggregation quality. More importantly, infrastructure centric coordination underuses a natural source of learning efficiency in vehicular environments, namely direct collaboration among vehicles that experience similar local contexts and can often exchange updates more cheaply than through a base station.

A more compelling substrate for collaboration is provided by hybrid 5G C-V2X networks, standardized by 3GPP [9]. In 3GPP compliant V2X systems, vehicles can communicate directly over the PC5 sidelink, operating in the 5.9 GHz ITS band, to exchange messages directly with nearby vehicles at low latency and minimal infrastructure cost [10]. When longer-range communication is required, the Uu interface is used through the 5G cellular network at higher cost but broader reach [11], thereby exposing two communication modes with markedly different latency, range, and resource profiles. This dual-interface capability creates a natural substrate for decentralized vehicle-to-vehicle (V2V) learning: nearby vehicles can collaborate efficiently and frequently share similar environment context, yet the most important collaborators may not always be the nearest ones. The PC5 sidelink can therefore support low latency exchange among adjacent vehicles, whereas the Uu interface can selectively reach statistically useful but

This paper was produced by the IEEE Publication Technology Group. They are in Piscataway, NJ.

Manuscript received April 19, 2021; revised August 16, 2021.

geographically distant collaborators through the cellular infrastructure. At the same time, local datasets are non-IID across vehicles, as each reflects a distinct driving environment. Effective collaboration must consequently be selective, personalized, and context aware.

Recent works in the collaborative learning literature [12]–[18] show that personalization and topology structure are key elements in achieving per client satisfactory models. Many approaches explored in the literature demonstrate that carefully selected collaborators can substantially improve vehicle specific performance under heterogeneous data and constrained communication. Yet the assumptions behind these approaches remain largely misaligned with vehicular networks. Most operate over comparatively stable client populations, infer collaboration in a logically centralized manner, or assume a single communication interface. Existing vehicular decentralized schemes exhibit the complementary limitation: they improve training efficiency under mobility and wireless constraints, but they do not jointly address personalization, hybrid link heterogeneity, or Byzantine robustness within a unified topology control problem [19]–[21]. In this setting, the collaboration topology is not a networking detail. It is a learning variable because the value of a collaborator depends jointly on vehicle similarity, environment context, link cost, and vehicle trustworthiness.

Vehicular neighborhoods are highly dynamic, as vehicles move rapidly across space and maintain only short-lived communication opportunities. Each vehicle must therefore learn under strict bandwidth, energy, and time constraints. Additionally, vehicular learning takes place in an open environment characterized by strong data heterogeneity and potentially unreliable participants. Personalization and Byzantine resilience are therefore essential requirements in V2X environments. Indiscriminate collaboration with many peers can waste scarce resources, increase computation overhead, and even degrade learning performance when selected neighbors are statistically irrelevant or malicious. These challenges motivate the central question of this paper: how should a vehicle in a hybrid 5G C-V2X network decide, at each stage of training, which neighbors to collaborate with and over which interface, so as to maximize personalized learning benefit under mobility, resource, and trust constraints without relying on centralized coordination?

We answer this question by proposing DANTE (Dual-interface Attention-guided Network-formation for Trustworthy vehicular Edge learning), a robust decentralized personalized learning framework built on dynamic network formation over a hybrid V2X collaboration graph. Unlike conventional vehicular FL, which treats communication as a mere transport layer, DANTE jointly models statistical relevance, radio reachability, and trust in topology control. Each vehicle maintains a personalized model and acts as a strategic agent that selects collaborators and assigns aggregation weights. The utility of each candidate link captures expected personalized loss reduction, communication and computation cost, and

trust-aware robustness. Short-range exchanges are carried over the PC5 sidelink, while the Uu interface is used selectively to reach high-value collaborators beyond direct radio range. To adapt to rapidly changing neighborhoods, each vehicle learns its link formation policy through a local attention-guided PPO mechanism that encodes candidate neighbors, link quality, budget state, and past collaboration quality, and activates only links expected to improve personalization per unit cost. Training is fully decentralized: no central critic, shared parameters, or global state is required, and each agent learns solely from local observations and local reward. Robust aggregation and online trust estimation are integrated into the same loop, allowing vehicles to mitigate suspicious updates and revert to local training when collaboration becomes unreliable.

The main contributions of this paper are summarized as follows.

- 1) We formulate robust decentralized personalized learning over hybrid 5G C-V2X as a directed network formation game in which vehicles are strategic players, link formation decisions define the action space, and payoffs integrate personalized loss improvement, interface aware resource cost, and trust weighted collaboration quality.
- 2) Each vehicle independently learns a Dante policy for neighbor selection and resource allocation from local observations and local reward, without any central critic, shared parameters, or global state, consistent with the operational constraints of V2X networks.
- 3) We evaluate DANTE against several baselines within a high fidelity SUMO driven simulation testbed that reproduces real world urban road networks across six locations, spanning dense urban grids, high speed highway corridors, coastal boulevards, and campus road networks, under realistic time varying mobility and V2X link dynamics.

II. Related Work

The growing demand for privacy-preserving, low-latency collaborative intelligence at the network edge has made decentralized learning a major research direction, with a large body of work now addressing serverless optimization, personalized collaboration, communication efficiency, and topology design. Early decentralized federated optimization focused on replacing server-based aggregation with peer-to-peer mixing. D-PSGD [22] established the canonical gossip-based training paradigm in which clients alternate local SGD and neighbor averaging over a fixed mixing graph. D² [26] strengthened this line by removing the dominant penalty induced by data heterogeneity, while SGP [27] extended decentralized training to directed and time-varying communication graphs through PushSum-based gossip. CHOCO-SGD [23] further reduced communication overhead by combining decentralized optimization with compressed message exchange. These methods provide the algorithmic foundation of decentralized federated learning, but they still optimize a shared objective

TABLE I

Comparison of DANTE with representative baselines across the key design dimensions of decentralized personalized learning in vehicular networks.

Method	Adaptive Topology	Personalized	Resource Aware	Dual Interface	Byzantine Robust	Learned Selection
FedAvg [4]	×	×	×	×	×	×
D-PSGD [22]	×	×	×	×	×	×
CHOCO-SGD [23]	×	×	✓	×	×	×
BALANCE [24]	×	×	×	×	✓	×
MDFL [19]	✓	×	✓	×	×	✓
CDPL [25]	✓	✓	×	×	✓	✓
DPFL [12]	✓	✓	✓	×	×	×
pFedGraph [14]	✓	✓	×	×	×	×
DANTE (ours)	✓	✓	✓	✓	✓	✓

over generic peer graphs and therefore do not address personalized collaboration.

This limitation becomes particularly restrictive in vehicular networks, where clients are mobile, communication opportunities are short-lived, and local data distributions are strongly context dependent. Barbieri et al. [28] demonstrated that consensus-driven decentralized federated learning can support extended sensing in connected vehicles through direct V2V model exchange. Yuan et al. [21] proposed peer-to-peer federated continual learning for naturalistic driving action recognition, showing that decentralized exchange can preserve privacy while mitigating catastrophic forgetting. Chen et al. [19] moved closer to practical V2X deployment by optimizing local iteration counts and leader selection under mobility and wireless constraints through multi-agent reinforcement learning. Although these studies bring decentralized learning into vehicular settings, they primarily target efficient coordination of shared-model training and do not jointly address personalization, dual link mode for efficient and low latency communication, and Byzantine-aware collaboration.

Because vehicular data are inherently non-IID, personalized decentralized learning is a more natural formulation than consensus on a single global model. Various works [25], [29], [30] explored the concept of personalized learning over peer-to-peer networks. [25] proposed CDPL, a robust and adaptive algorithm to learn personalized models in dynamic networks by evaluating the received updates against a small subset of the peer’s local dataset. The authors in [18] suggested to jointly learn personalized models and sparse collaboration graphs in a fully decentralized setting. DisPFL [31] introduced decentralized sparse training and personalized masks, reducing both communication and computation while maintaining client-specific models. DPFL [12], which we also implement, cast decentralized personalization as a bilevel optimization problem and used constrained greedy graph construction to select collaborators under a fixed budget. These methods establish that who collaborates with whom is central to personalization quality, yet they do not account for the rapidly changing neighborhoods, asymmetric interface costs, and trust uncertainty that define vehicular V2X

environments.

Operational constraints add a second layer of difficulty. Communication and computation efficiency have been studied through compression, resource allocation, and graph sparsification. CHOCO-SGD [23] reduces payload size through compressed gossip, while Meng et al. [20] jointly optimize neighbor selection, transmit power, and CPU frequency using graph neural networks in decentralized wireless learning. Zhang et al. [32] showed that graph sparsification can substantially reduce energy consumption at bottleneck nodes while preserving learning quality. Closely related to these efforts is the topology-optimization literature, which treats the collaboration graph itself as a statistical design variable. pFedGraph [14] infers collaboration weights from model similarity and dataset size to produce client-specific aggregation patterns. DFedPGP [13] showed that directed collaboration can outperform symmetric peer exchange in personalized decentralized learning. Le Bars et al. [33] further demonstrated that topology should be adapted to data heterogeneity itself, and formalized topology learning as an optimization problem driven by neighborhood heterogeneity. Together, these works show that resource allocation and topology design are decisive for decentralized learning performance, but they stop short of unifying personalization, hybrid V2X link heterogeneity, and adversarial robustness in a single decentralized control framework.

DANTE differs from all of these approaches in three coupled ways. First, it formulates vehicular collaboration as a dynamic network formation problem rather than fixed gossip, centralized graph inference, or static budgeted graph construction. Second, it explicitly differentiates the cost and reachability of PC5 sidelink and Uu relay communication, allowing each vehicle to trade off local low-cost exchange against longer-range but more expensive collaboration. Third, it embeds trust estimation and robust aggregation directly into topology control, so collaborator selection depends jointly on personalized learning value, communication cost, and reliability. To the best of our knowledge, no prior method combines these properties in a fully decentralized personalized learning framework for hybrid 5G C-V2X networks.

III. System Model and Problem Definition

We consider a fully decentralized vehicular learning system in which connected vehicles collaboratively train personalized models over a hybrid 5G C-V2X network. Each vehicle aims to learn a model adapted to its own local context while selectively incorporating updates from statistically compatible vehicles. Collaboration is performed without any central coordination and relies on both PC5 sidelink and 5G Uu communication links, which exhibit different range, latency, and cost characteristics. In the following, we formalize the network topology enabling personalized training and its associated computation and communication cost model.

A. Collaboration Graph

We consider a time-varying vehicular network observed over discrete decision rounds $t \in \{0, \dots, T-1\}$. At each round t , let $\mathcal{V}_t$ denote the set of active vehicles participating in collaborative learning. Since vehicle locations and wireless connectivity evolve continuously, the communication topology is inherently dynamic. Model exchange is supported by a hybrid V2X network with two communication modes: direct PC5 sidelink and infrastructure-assisted 5G Uu relay. We represent feasible communication opportunities by the graph

$$\mathcal{G}_t^{\text{feas}} = (\mathcal{V}_t, \mathcal{E}_t^{\text{sl}}, \mathcal{E}_t^{\text{uu}}),$$

where $(i, j) \in \mathcal{E}_t^{\text{sl}}$ if a direct PC5 transmission between vehicles i and j is feasible at round t , and $(i, j) \in \mathcal{E}_t^{\text{uu}}$ if a 5G Uu path through the infrastructure is available. For each pair (i, j) , we define the feasible communication mode set as

$$\mathcal{M}_{ij}^{(t)} = \{m \in \{\text{sl}, \text{uu}\} : (i, j) \in \mathcal{E}_t^m\}.$$

The feasible neighbor set of vehicle i is then

$$\mathcal{F}_i^{(t)} = \{j \in \mathcal{V}_t \setminus \{i\} : \mathcal{M}_{ij}^{(t)} \neq \emptyset\}.$$

The actual collaboration is modeled by the directed graph

$$\mathcal{G}_t^{\text{coll}} = (\mathcal{V}_t, \mathcal{E}_t^{\text{coll}}),$$

where $(j \rightarrow i) \in \mathcal{E}_t^{\text{coll}}$ indicates that vehicle i uses the update received from vehicle j at round t . The collaboration neighborhood of vehicle i is therefore

$$C_i^{(t)} = \{j \in \mathcal{F}_i^{(t)} : (j \rightarrow i) \in \mathcal{E}_t^{\text{coll}}\}.$$

For each active collaboration edge, vehicle i selects a communication mode $m_{ij}^{(t)} \in \mathcal{M}_{ij}^{(t)}$ and assigns a collaboration weight $\alpha_{ij}^{(t)} \geq 0$, subject to

$$\sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} \leq 1, \quad \alpha_{ii}^{(t)} = 1 - \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)}.$$

Hence, vehicle i always retains a fraction of its own model while selectively incorporating updates from its collaborators. The collaboration graph is asymmetric and overlapping, that is, $j \in C_i^{(t)}$ does not imply $i \in C_j^{(t)}$.

B. Learning Model

Each vehicle $i \in \mathcal{V}_t$ holds a private dataset $\mathcal{D}_i = \{(x_{i,k}, y_{i,k})\}_{k=1}^{n_i}$ drawn from a local distribution P_i . The distributions $\{P_i\}$ are heterogeneous across vehicles, yielding non-IID data. Vehicle i maintains a personalized model $w_i^{(t)} \in \mathbb{R}^d$ and aims to minimize its population risk

$$F_i(w) = \mathbb{E}_{(x,y) \sim P_i} [\ell(w; x, y)].$$

In practice, since P_i is unknown, local optimization is performed on the empirical risk

$$\hat{F}_i(w) = \frac{1}{n_i} \sum_{k=1}^{n_i} \ell(w; x_{i,k}, y_{i,k}).$$

The personalized objective of vehicle i is therefore

$$\min_{w_i} F_i(w_i).$$

Since $P_i \neq P_j$ in general, enforcing $w_i = w_j$ for all vehicles is suboptimal. At the system level, the objective is to learn the set of personalized models $W = \{w_i\}_{i \in \mathcal{V}_t}$ by minimizing the aggregate multi-task loss

$$\min_W \mathcal{L}(W) = \sum_{i \in \mathcal{V}_t} \rho_i F_i(w_i), \quad \rho_i \geq 0, \quad \sum_{i \in \mathcal{V}_t} \rho_i = 1,$$

where ρ_i denotes the weight of vehicle i . Unlike classical Decentralized Learning (DL), this objective does not impose model consensus.

At the beginning of round t , vehicle i receives the most recent models $\hat{w}_j^{(t)}$ from its collaborators $j \in C_i^{(t)}$ and forms the aggregate model $\bar{w}_i^{(t)}$.

$$\bar{w}_i^{(t)} = \alpha_{ii}^{(t)} w_i^{(t)} + \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} \hat{w}_j^{(t)}, \quad (1)$$

where $\hat{w}_j^{(t)}$ is the model received from vehicle j , and

$$\alpha_{ii}^{(t)} + \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} = 1, \quad \alpha_{ij}^{(t)} \geq 0.$$

Starting from $\bar{w}_i^{(t)}$, vehicle i performs $I_i^{(t)}$ local SGD steps on mini-batches of size b_i sampled from $\mathcal{D}_i$:

$$w_i^{(t, \ell+1)} = w_i^{(t, \ell)} - \eta_i^{(t)} g_i(w_i^{(t, \ell)}; \xi_i^{(t, \ell)}), \quad (2)$$

for $\ell = 0, \dots, I_i^{(t)} - 1$, with $w_i^{(t, 0)} = \bar{w}_i^{(t)}$, where $\xi_i^{(t, \ell)}$ is a mini-batch of size b_i drawn from $\mathcal{D}_i$ and $g_i(\cdot; \xi_i^{(t, \ell)})$ is the corresponding stochastic gradient of $\hat{F}_i$. The updated personalized model is then

$$w_i^{(t+1)} = w_i^{(t, I_i^{(t)})}.$$

C. Computation and Communication Model

Each vehicle incurs both local computation cost and V2X communication cost during collaborative training. We model these costs at the granularity of one learning round.

If vehicle i performs $I_i^{(t)}$ local SGD steps on mini-batches of size b_i , with c_i CPU cycles per sample and

processor frequency $f_i^{(t)}$, its local computation time and energy are

$$\tau_{i,\text{cmp}}^{(t)} = \frac{I_i^{(t)} b_i c_i}{f_i^{(t)}}, \quad E_{i,\text{cmp}}^{(t)} = \kappa I_i^{(t)} b_i c_i (f_i^{(t)})^2$$

where κ is the effective switched capacitance.

Vehicles exchange model updates over hybrid 5G C-V2X links, either through direct PC5 sidelink or infrastructure-assisted 5G Uu relay, which exhibit different range, latency, and power characteristics [9]–[11]. Let S denote the model payload size in bits. For a PC5 sidelink transmission from vehicle j to vehicle i , the achievable rate is

$$R_{ji,\text{sl}}^{(t)} = B_{ji,\text{sl}}^{(t)} \log_2(1 + \text{SINR}_{ji,\text{sl}}^{(t)}),$$

with transmission time and energy

$$\tau_{ji,\text{sl}}^{(t)} = \frac{S}{R_{ji,\text{sl}}^{(t)}}, \quad E_{ji,\text{sl}}^{(t)} = p_{j,\text{sl}}^{(t)} \tau_{ji,\text{sl}}^{(t)}.$$

For a 5G Uu relay path through the serving gNB, the end-to-end transfer time is determined by the uplink and downlink legs:

$$\tau_{ji,\text{uu}}^{(t)} = \frac{S}{R_{j \rightarrow g}^{u,(t)}} + \frac{S}{R_{g \rightarrow i}^{d,(t)}},$$

and the associated communication energy is

$$E_{ji,\text{uu}}^{(t)} = p_j^{u,(t)} \frac{S}{R_{j \rightarrow g}^{u,(t)}} + p_i^{\text{rx},(t)} \frac{S}{R_{g \rightarrow i}^{d,(t)}}.$$

This model captures the core asymmetry of hybrid V2X collaboration: PC5 is typically cheaper and lower-latency but range-limited, whereas 5G Uu relay offers larger reach at higher communication cost.

D. Threat Model

This work considers an open vehicular environment in which a subset $\mathcal{B}_t \subseteq \mathcal{V}_t$ may behave in a Byzantine manner at round t . A Byzantine vehicle is assumed to know the protocol, collude with other Byzantine vehicles, and send arbitrary, inconsistent, or strategically poisoned model updates, but it cannot directly compromise the local computation of honest vehicles. The threat model covers crash and communication failures, free-riding or lazy participants that consume updates without contributing useful models, and adversarial model poisoning aimed at introducing backdoors, causing divergence, or steering learning toward ineffective solutions. We particularly consider the case where Byzantine vehicles collude to dominate the neighborhood of an honest vehicle and corrupt its personalized model. To mitigate the effect of unreliable or Byzantine neighbors, each vehicle maintains a local trust score $q_{ij}^{(t)} \in [0, 1]$ for every feasible neighbor $j \in \mathcal{F}_i^{(t)}$. This score represents vehicle i 's current estimate of the reliability of j 's transmitted model updates.

IV. DANTE: Dynamic Network Formation and Decentralized Solution

Having defined the system model in Section III, we now answer the central question of this work: how should each vehicle decide, at every round, with whom to collaborate so as to maximize its personalized learning benefit under the energy, bandwidth, latency, and trust constraints of a hybrid 5G C-V2X network? We proceed in two steps. First, we cast vehicular decentralized personalized learning (VDPL) as a dynamic network formation game played over the time-varying collaboration graph, in which each vehicle is a strategic player that forms and severs directed links to its peers. Second, we introduce Dante (Dual-interface Attention-guided Network-formation for Trustworthy vehicular Edge learning), a fully decentralized solver in which every vehicle learns its own link-formation policy via an attention-guided PPO actor-critic, aggregates the received models through a fused attention-trust-robustness weighting, and maintains a local trust-update loop. No central critic, parameter server, or globally observed state is required at any stage.

A. Dynamic Network Formation as a Strategic Markov Game

We model collaboration in VDPL as a discrete-time strategic network formation game in the tradition of Jackson and Wolinsky [34]–[36]. This class of models treats agents as deciding which directed links to form or sever and lets payoffs depend on the resulting graph rather than on a partition of the agent set into disjoint coalitions. This graph-level abstraction is well suited to vehicular collaboration, where links are overlapping, directed, and asymmetric: vehicle i may benefit from j 's update without any reciprocity obligation, and each vehicle observes its environment only locally.

Definition 1 (Dynamic network formation game for VDPL). A dynamic network formation game for VDPL is the tuple

$$\mathfrak{G} = \left(\{\mathcal{V}_t\}_{t \geq 0}, \mathcal{S}, \{\mathcal{A}_i(\cdot)\}_i, \mathcal{P}, \{u_i\}_i, \gamma \right),$$

whose components are defined as follows.

- **Players.** At round t , the players are the active vehicles $\mathcal{V}_t$ of Section III. Every honest vehicle is a strategic, self-interested agent that maximizes its own personalized learning performance under local resource constraints.
- **State and local observation.** The global state $s^{(t)} \in \mathcal{S}$ collects vehicle positions, channel conditions, resource budgets $b_i^{(t)}$, personalized models $\{w_i^{(t)}\}$, and trust vectors $\{q_{ij}^{(t)}\}$. No vehicle observes $s^{(t)}$. Vehicle i receives only a local observation $o_i^{(t)} \in \mathcal{O}_i$ that exposes its own state together with per-candidate link features $\{f_{ij}^{(t)}\}$ and trust scores $\{q_{ij}^{(t)}\}$ restricted to the feasible neighbor set $\mathcal{F}_i^{(t)}$ defined in Section III.

- Feasible actions. Vehicle i plays a per-candidate binary vector

$$a_i^{(t)} = (a_{ij}^{(t)})_{j \in \mathcal{F}_i^{(t)}} \in \{0, 1\}^{|\mathcal{F}_i^{(t)}|},$$

where $a_{ij}^{(t)} = 1$ accepts the model update of neighbor j at round t . Actions induce the collaboration set $C_i^{(t)} = \{j \in \mathcal{F}_i^{(t)} : a_{ij}^{(t)} = 1\}$ and the directed collaboration graph $\mathcal{G}_t^{\text{coll}}$ of Section III, with $(j \rightarrow i) \in \mathcal{E}_t^{\text{coll}} \iff a_{ij}^{(t)} = 1$. Since every candidate $j \in \mathcal{F}_i^{(t)}$ is already tagged with its physical interface $m_{ij}^{(t)} \in \mathcal{M}_{ij}^{(t)}$ by the communication layer, vehicle i chooses directed edges rather than separately optimizing the interface of a fixed peer.

- Per-round payoff. Vehicle i receives the instantaneous utility

$$u_i^{(t)} = \Delta_i^{(t)} - \lambda_E E_i^{(t)} - \lambda_B B_i^{(t)} - \lambda_T T_i^{(t)}, \quad (3)$$

where

$$\Delta_i^{(t)} = \widehat{F}_{i,\text{val}}(w_i^{(t)}) - \widehat{F}_{i,\text{val}}(w_i^{(t+1)})$$

is the loss reduction measured on a held-out local validation split $\mathcal{D}_i^{\text{val}} \subset \mathcal{D}_i$, $E_i^{(t)}$, $B_i^{(t)}$, and $T_i^{(t)}$ are the round-level energy, bandwidth, and latency costs inherited from the computation and communication model of Section III, and $\lambda_E, \lambda_B, \lambda_T \geq 0$ are per-vehicle cost weights.

- Transitions. The state evolves according to a kernel $\mathcal{P}(\cdot \mid s^{(t)}, a^{(t)})$ jointly induced by vehicle mobility, wireless-channel stochasticity, resource depletion $b_i^{(t+1)} = b_i^{(t)} - [E_i^{(t)}, B_i^{(t)}]^\top$, trust evolution (Section IV-C2), and the personalized model update (Section IV-C). Vehicle i does not observe $\mathcal{P}$ and interacts with it only through $o_i^{(t)}$.

Long-term objective. Given a joint policy profile $\pi = (\pi_i)_i$, vehicle i seeks a local policy that maximizes its discounted infinite-horizon return

$$J_i(\pi_i; \pi_{-i}) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t u_i^{(t)} \right], \quad \gamma \in (0, 1), \quad (4)$$

where π_{-i} denotes the policy profile of the remaining vehicles. Because every vehicle optimizes its own return conditional on the strategies of the others, Equation (4) defines a Markov game over the evolving collaboration graph rather than a joint cooperative objective; we return to the Nash-equilibrium question in Section IV-D.

Constraints. At each round, the decision of vehicle i must additionally satisfy

$$|C_i^{(t)}| \leq K_i, \quad (5)$$

$$E_{i,\text{cmp}}^{(t)} + \sum_{j \in C_i^{(t)}} E_{ij}^{m_{ij},(t)} \leq \bar{E}_i^{(t)}, \quad (6)$$

$$\sum_{j \in C_i^{(t)}} B_{ij}^{m_{ij},(t)} \leq \bar{B}_i^{(t)}, \quad (7)$$

$$T_{i,\text{cmp}}^{(t)} + \max_{j \in C_i^{(t)}} T_{ij}^{m_{ij},(t)} \leq \bar{T}_i^{(t)}, \quad (8)$$

where K_i is the per-vehicle neighborhood cap and $\bar{E}_i^{(t)}, \bar{B}_i^{(t)}, \bar{T}_i^{(t)}$ are the per-round energy, bandwidth, and latency budgets of Section III. Latency (8) and neighborhood size (5) are hard per-round caps enforced at action-admissibility time, whereas the energy and bandwidth caps (6)–(7) are tracked through the budget state $b_i^{(t)}$ so that a vehicle that overspends early loses collaboration capacity later. This design keeps the problem amenable to policy-gradient methods without resorting to a Lagrangian relaxation, while still penalizing budget depletion endogenously through the return.

B. Utility of Collaboration

The stage payoff (3) uses the validation-loss drop $\Delta_i^{(t)}$ as its observable learning signal. That quantity is suitable for online decision making but only reveals the value of collaboration after a round has been completed. To expose the learning geometry underlying $\Delta_i^{(t)}$ and to support the theoretical analysis, we introduce an analytical surrogate based on gradient agreement.

Let $g_i^{(t)}$ and $g_j^{(t)}$ denote stochastic gradients of vehicles i and j evaluated at their current iterates. We measure directional agreement by the cosine similarity

$$s_{ij}^{(t)} = \frac{(g_i^{(t)})^\top g_j^{(t)}}{\|g_i^{(t)}\| \|g_j^{(t)}\| + \varepsilon} \in [-1, 1],$$

with $\varepsilon > 0$ a numerical-stability constant. This quantity admits a first-order interpretation: Taylor-expanding F_i around $w_i^{(t)}$ gives

$$F_i(w_i^{(t)} - \eta g_j^{(t)}) = F_i(w_i^{(t)}) - \eta (\nabla F_i(w_i^{(t)}))^\top g_j^{(t)} + O(\eta^2),$$

so under the standard assumption that $g_i^{(t)}$ is unbiased for $\nabla F_i(w_i^{(t)})$, a positive $s_{ij}^{(t)}$ indicates, in expectation, that j 's update direction is aligned with the local descent direction of i . Combining alignment with the trust score $q_{ij}^{(t)} \in [0, 1]$ introduced in Section IV-C2, we define the trust-weighted collaboration surrogate

$$G_i(C_i^{(t)}, s^{(t)}) = \sum_{j \in C_i^{(t)}} q_{ij}^{(t)} s_{ij}^{(t)}. \quad (9)$$

Large values of G_i correspond to collaboration sets whose update directions are, on average, both aligned with the local descent of i and originating from trusted peers, whereas negative values indicate that the selected neighbors tend to push the model away from descent. The surrogate is introduced for interpretation and convergence analysis (Section IV-D); the reward that drives the policy is the observable $\Delta_i^{(t)}$ entering (3).

C. Dante: Attention-Guided Decentralized Solver

Dante factorizes the per-vehicle policy through three coupled components: (i) a graph-attention policy head that scores candidate peers from local link features, (ii) a robust personalized aggregation layer that fuses attention with trust and a coordinate-wise robustness filter, and

(iii) a local trust update that closes the loop by down-weighting peers whose past contributions hurt the local loss. Every component runs locally on vehicle i using $o_i^{(t)}$ alone; no joint state, joint critic, or shared parameters are ever assembled.

1) GAT policy parameterization: Let $x_i^{(t)} \in \mathbb{R}^{d_o}$ be the ego feature vector of vehicle i and $x_{ij}^{(t)} \in \mathbb{R}^{d_n}$ the feature vector of candidate neighbor $j \in \mathcal{F}_i^{(t)}$. A single-head graph-attention encoder computes

$$\begin{aligned} h_i^{(t)} &= W_s x_i^{(t)}, & v_{ij}^{(t)} &= W_n x_{ij}^{(t)}, \\ \beta_{ij}^{(t)} &= \frac{\exp(e_{ij}^{(t)})}{\sum_{k \in \mathcal{F}_i^{(t)}} \exp(e_{ik}^{(t)})}, \end{aligned} \quad (10)$$

with unnormalized logits $e_{ij}^{(t)} = a^\top \text{LeakyReLU}([h_i^{(t)} \parallel v_{ij}^{(t)}])$ and learnable parameters W_s , W_n , and a . The neighborhood context is $c_i^{(t)} = \sum_{j \in \mathcal{F}_i^{(t)}} \beta_{ij}^{(t)} v_{ij}^{(t)}$. The raw attention $\beta_{ij}^{(t)}$ plays a dual role: it drives the actor's Bernoulli logits, and, combined with the trust and robustness scores introduced below, it shapes the final aggregation weights $\alpha_{ij}^{(t)}$. This coupling ties whom the policy chooses to listen to directly to how much weight that peer carries in the model update.

Conditioned on $(h_i^{(t)}, c_i^{(t)}, v_{ij}^{(t)})$, the actor outputs a per-candidate Bernoulli and the critic outputs a scalar value:

$$\begin{aligned} \pi_{\theta_i}(a_{ij}^{(t)} = 1 \mid o_i^{(t)}) &= \sigma(f_{\theta_i}([h_i^{(t)} \parallel c_i^{(t)} \parallel v_{ij}^{(t)}])), \\ V_{\phi_i}(o_i^{(t)}) &= g_{\phi_i}([h_i^{(t)} \parallel c_i^{(t)}]), \end{aligned} \quad (11)$$

where σ is the logistic sigmoid. Because the per-candidate actions are conditionally independent Bernoullis, the joint log-probability factorizes as $\log \pi_{\theta_i}(a_i^{(t)} \mid o_i^{(t)}) = \sum_{j \in \mathcal{F}_i^{(t)}} \log \pi_{\theta_i}(a_{ij}^{(t)} \mid o_i^{(t)})$, making per-vehicle PPO updates tractable regardless of $|\mathcal{F}_i^{(t)}|$. Each vehicle stores only its own parameters (θ_i, ϕ_i) : no critic ever observes a joint state or joint action, and the scheme avoids the realizability issues of centralized-training-with-decentralized-execution schemes in V2X, where a joint state would presuppose a central entity that our setting precludes. If more than K_i candidates are sampled as active, Dante retains the K_i highest-attention edges, which enforces the neighborhood cap (5) by construction.

2) Trust score and robustness filter: Vehicle i maintains a local trust score $q_{ij}^{(t)} \in [0, 1]$ for every feasible neighbor $j \in \mathcal{F}_i^{(t)}$, interpreted as vehicle i 's current estimate of the reliability of j 's transmitted updates. The score is updated by an exponentially weighted recursion

$$q_{ij}^{(t+1)} = (1 - \rho) q_{ij}^{(t)} + \rho \phi_{ij}^{(t)}, \quad \rho \in (0, 1], \quad (12)$$

where the per-round reliability signal $\phi_{ij}^{(t)} \in [0, 1]$ is defined as the product of two local indicators,

$\phi_{ij}^{(t)} = \mathbf{1}\{r_{ij}^{(t)} \geq r_{\min}\} \cdot \mathbf{1}\{\widehat{F}_{i,\text{val}}(w_i^{(t+1)}) \leq \widehat{F}_{i,\text{val}}(w_i^{(t)}) + \tau_{\text{val}}\}$, namely (i) vehicle j passed the local robust filter at round t and (ii) the validation loss of vehicle i did not increase after including j 's update, up to a small slack $\tau_{\text{val}} \geq 0$. Because

$\phi_{ij}^{(t)} \in [0, 1]$ and $\rho \in (0, 1]$, the recursion keeps $q_{ij}^{(t)} \in [0, 1]$ for every t . The smoothing in (12) gives persistence against transient link errors while still allowing a sustained Byzantine peer to be progressively de-weighted.

The robustness score $r_{ij}^{(t)} \in [0, 1]$ is produced by a coordinate-wise robust filter applied to the set of received models $\{\hat{w}_j^{(t)}\}_{j \in C_i^{(t)}}$. Let $\text{TM}_i^{(t)}$ denote the coordinate-wise trimmed mean of $\{\hat{w}_j^{(t)}\}_{j \in C_i^{(t)}}$ and let $d_{ij}^{(t)} = \|\hat{w}_j^{(t)} - \text{TM}_i^{(t)}\|_2$; we set $r_{ij}^{(t)} = \exp(-d_{ij}^{(t)}/\sigma_r) \in (0, 1]$, so that peers that drift far from the trimmed-mean consensus contribute near-zero weight while still-close peers retain near-unit score. Coordinate-wise trimmed mean is a standard robust aggregator in federated learning and tolerates up to a fraction $f/|C_i^{(t)}| < 1/2$ of Byzantine neighbors with bounded honest gradients [37], [38].

3) Fused attention-trust-robustness aggregation: Dante combines the GAT attention $\beta_{ij}^{(t)}$, the trust score $q_{ij}^{(t)}$, and the robustness score $r_{ij}^{(t)}$ into the final aggregation weight that instantiates the collaborative update of Section III-B:

$$\alpha_{ij}^{(t)} = (1 - \alpha_{ii}^{(t)}) \cdot \frac{\beta_{ij}^{(t)} q_{ij}^{(t)} r_{ij}^{(t)}}{\varepsilon + \sum_{k \in C_i^{(t)}} \beta_{ik}^{(t)} q_{ik}^{(t)} r_{ik}^{(t)}}, \quad j \in C_i^{(t)}, \quad (13)$$

with self-retention weight $\alpha_{ii}^{(t)} \in (0, 1)$ controlling the personalization/collaboration trade-off. By construction,

$$\alpha_{ii}^{(t)} + \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} = 1, \quad \alpha_{ij}^{(t)} \geq 0,$$

so (13) satisfies the convex-combination constraint of Section III-B and can be substituted directly into the aggregation rule of that section, yielding $\bar{w}_i^{(t)} = \alpha_{ii}^{(t)} w_i^{(t)} + \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} \hat{w}_j^{(t)}$. Starting from $\bar{w}_i^{(t)}$, vehicle i then executes $I_i^{(t)}$ local SGD steps as prescribed in Section III-B to obtain the next personalized iterate $w_i^{(t+1)}$. The round ordering is therefore aggregate-then-train, consistent with the system model.

The fusion in (13) is also a graceful fallback. If every neighbor in $C_i^{(t)}$ is Byzantine, the robust filter and trust recursion drive $q_{ij}^{(t)} r_{ij}^{(t)} \rightarrow 0$, so $\alpha_{ij}^{(t)} \rightarrow 0$ and $\bar{w}_i^{(t)} \rightarrow w_i^{(t)}$; the collaborative update degenerates to pure local SGD, which is the safest behavior available to an honest vehicle.

4) Local PPO optimization: Each vehicle updates its actor-critic pair by Proximal Policy Optimization on local trajectories only. Let $\delta_i^{(t)} = u_i^{(t)} + \gamma V_{\phi_i}(o_i^{(t+1)}) - V_{\phi_i}(o_i^{(t)})$ denote the temporal-difference error and $\hat{A}_i^{(t)} = \sum_{h \geq 0} (\gamma \lambda_{\text{GAE}})^h \delta_i^{(t+h)}$ the generalized advantage estimate. The clipped PPO surrogate with entropy regularization is

$$\begin{aligned} \mathcal{L}_i^{\text{PPO}}(\theta_i) &= \mathbb{E} \left[\min(\rho_i^{(t)} \hat{A}_i^{(t)}, \text{clip}(\rho_i^{(t)}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i^{(t)}) \right] \\ &\quad + \eta_{\text{ent}} \mathbb{E} [\mathcal{H}(\pi_{\theta_i}(\cdot \mid o_i^{(t)}))], \end{aligned} \quad (14)$$

with likelihood ratio $\rho_i^{(t)} = \pi_{\theta_i}(a_i^{(t)} \mid o_i^{(t)}) / \pi_{\theta_i^{\text{old}}}(a_i^{(t)} \mid o_i^{(t)})$, clip range $\epsilon \in (0, 1)$, entropy coefficient $\eta_{\text{ent}} \geq 0$, and

Shannon entropy $\mathcal{H}$. The critic regresses on the return targets $\hat{R}_i^{(t)} = \hat{A}_i^{(t)} + V_{\phi_i}(o_i^{(t)})$. Each vehicle therefore runs an independent local PPO loop in the style of IPPO [39], without sharing trajectories, critics, or parameters with its peers.

5) The DANTE algorithm: Algorithm 1 summarizes the complete per-vehicle procedure. Each iteration performs one round of environment interaction followed by local learning, and the PPO update is triggered every K_{upd} rounds from the local rollout buffer. All steps operate on local data and locally exchanged parameters; no global reward, joint state, or joint critic is ever assembled.

D. Convergence and Complexity

The joint problem is non-convex and non-stationary, and global optimality of coupled model and policy updates is out of reach. Following a two-timescale argument standard in multi-agent reinforcement learning, we analyze the learning and policy layers separately: convergence of the personalized learning iterates under a fixed collaboration profile, and stationarity of the local PPO update under an asymptotically stationary local MDP. Detailed proofs are deferred to Appendix A.

a) Assumptions.: We use the following standard conditions.

- A1. (Smoothness) Each local objective F_i is L -smooth and bounded below by F_i^{inf} .
- A2. (Unbiased bounded-variance gradients) $\mathbb{E}[g_i(w; \xi)] = \nabla F_i(w)$ and $\mathbb{E}\|g_i(w; \xi) - \nabla F_i(w)\|^2 \leq \sigma^2$.
- A3. (Robust-aggregation bias) The fused weights $\{\alpha_{ij}^{(t)}\}$ of (13) together with the robust filter yield an effective post-aggregation stochastic gradient $\tilde{g}_i^{(t)}$ satisfying $\|\mathbb{E}[\tilde{g}_i^{(t)} | w_i^{(t)}] - \nabla F_i(w_i^{(t)})\| \leq \zeta$ uniformly in t , for some $\zeta \geq 0$.
- A4. (Bounded rewards and features) $|u_i^{(t)}| \leq u_{\text{max}}$ almost surely, and candidate features and attention outputs are uniformly bounded.
- A5. (Persistent exploration) The entropy bonus in (14) keeps $\pi_{\theta_i}(\cdot | o_i^{(t)})$ strictly positive on every action in the support of an ϵ -perturbation of the current policy.

A1–A2 are standard for smooth non-convex SGD [40]. A3 is the standard interface used in robust federated learning to decouple filter geometry from convergence arguments [37], [38]; the coordinate-wise trimmed mean of Section IV-C2 attains $\zeta = \mathcal{O}(f/|C_i^{(t)}|) \cdot B$ for f Byzantine neighbors and bounded honest gradients $\|g_j\| \leq B$. A4 follows from the bounded validation-loss drop together with the per-round resource caps (6)–(8). A5 is enforced by the entropy term in (14).

b) Convergence of the learning layer under fixed policies.:

Proposition 1 (Robust personalized SGD convergence). Fix any joint policy profile π for which the induced collaboration sequence $\{C_i^{(t)}\}_{t \geq 0}$ is measurable. Under A1–A3, with diminishing step size $\eta_i^{(t)} = \eta_0/\sqrt{t+1}$ and

Algorithm 1 Dante: per-vehicle decentralized network formation and personalized learning, run locally by each vehicle i .

Input: local dataset $\mathcal{D}_i$; initial model $w_i^{(0)}$; initial trust $q_{ij}^{(0)} = 1$; actor–critic parameters (θ_i, ϕ_i) ; per-round budgets $(\bar{E}_i, \bar{B}_i, \bar{T}_i)$; neighborhood cap K_i ; PPO hyperparameters $(\gamma, \lambda_{\text{GAE}}, \epsilon, \eta_{\text{ent}}, K_{\text{upd}})$; self-weight schedule $\alpha_{ii}^{(t)}$.

Output: personalized model $w_i^{(T)}$ and policy (θ_i, ϕ_i) .

- 1: for $t = 0, 1, \dots, T-1$ do
- 2: Discover feasible peers $\mathcal{F}_i^{(t)}$ over PC5 sidelink and Uu relay; observe link features $\{f_{ij}^{(t)}\}$ and trust scores $\{q_{ij}^{(t)}\}$.
- 3: Encode ego and candidate features $(h_i^{(t)}, \{v_{ij}^{(t)}\})$; compute attention $\{\beta_{ij}^{(t)}\}$ and value $V_{\phi_i}(o_i^{(t)})$. $\triangleright$ Eq. (10)
- 4: Sample per-candidate Bernoulli actions $a_{ij}^{(t)} \sim \pi_{\theta_i}(\cdot | o_i^{(t)})$; form $C_i^{(t)} = \{j : a_{ij}^{(t)} = 1\}$; if $|C_i^{(t)}| > K_i$, retain the top- K_i edges ranked by $\beta_{ij}^{(t)}$. $\triangleright$ Eq. (11)
- 5: Exchange models with $C_i^{(t)}$ over the selected interface $m_{ij}^{(t)}$; receive $\{\hat{w}_j^{(t)}\}_{j \in C_i^{(t)}}$.
- 6: Apply the coordinate-wise robust filter to obtain $\{r_{ij}^{(t)}\}_{j \in C_i^{(t)}}$.
- 7: Form fused weights $\{\alpha_{ij}^{(t)}\}$ and aggregate $\bar{w}_i^{(t)} = \alpha_{ii}^{(t)} w_i^{(t)} + \sum_{j \in C_i^{(t)}} \alpha_{ij}^{(t)} \hat{w}_j^{(t)}$. $\triangleright$ Eq. (13)
- 8: Starting from $\bar{w}_i^{(t)}$, run $I_i^{(t)}$ local SGD steps to obtain $w_i^{(t+1)}$. $\triangleright$ Section III-B
- 9: Measure $(E_i^{(t)}, B_i^{(t)}, T_i^{(t)})$ and validation drop $\Delta_i^{(t)}$; form reward $u_i^{(t)}$. $\triangleright$ Eq. (3)
- 10: Evaluate $\phi_{ij}^{(t)}$ from the robust-filter indicator and the post-update validation loss; update trust $\{q_{ij}^{(t+1)}\}$. $\triangleright$ Eq. (12)
- 11: Append transition $(o_i^{(t)}, a_i^{(t)}, u_i^{(t)}, o_i^{(t+1)}, V_{\phi_i}(o_i^{(t)}))$ to local rollout buffer $\mathcal{R}_i$.
- 12: if $|\mathcal{R}_i| \geq K_{\text{upd}}$ then
- 13: Compute $\{\hat{A}_i^{(t)}, \hat{R}_i^{(t)}\}$ via generalized advantage estimation.
- 14: Update $\theta_i \leftarrow \theta_i + \eta_{\pi} \nabla_{\theta_i} \mathcal{L}_i^{\text{PPO}}(\theta_i)$; update ϕ_i by MSE on $\{\hat{R}_i^{(t)}\}$. $\triangleright$ Eq. (14)
- 15: Clear $\mathcal{R}_i$.
- 16: end if
- 17: end for

$\eta_0 \leq 1/L$, the personalized iterates $\{w_i^{(t)}\}$ generated by aggregate-then-train starting from $\bar{w}_i^{(t)}$ (Section III-B) satisfy, for every honest vehicle i ,

$$\begin{aligned} & \min_{0 \leq t \leq T-1} \mathbb{E}[\|\nabla F_i(w_i^{(t)})\|^2] \\ & \leq \frac{2(F_i(w_i^{(0)}) - F_i^{\text{inf}})}{\eta_0 \sqrt{T}} + \frac{L\eta_0 \sigma^2}{\sqrt{T}} + 2\zeta^2. \end{aligned} \quad (15)$$

Proposition 1 shows that the learning layer of Dante inherits the $\mathcal{O}(T^{-1/2})$ rate of non-convex SGD with an

additive $\mathcal{O}(\zeta^2)$ floor set by the worst-case aggregation bias under the fused weights (13). The proof applies the standard non-convex SGD descent lemma to the equivalent stochastic gradient induced by aggregation; see Appendix A.

Remark 1 (Byzantine fallback and collaboration gain). When every neighbor of vehicle i is honest and the trust and robustness scores are non-degenerate, A3 holds with $\zeta = 0$ and the floor vanishes. When every neighbor is Byzantine, $\alpha_{ij}^{(t)} \rightarrow 0$ by (13), the update reduces to pure local SGD, and the bound still holds with $\zeta = 0$; Dante therefore degrades gracefully to safe local training. Between these extremes, collaboration strictly helps vehicle i iff the effective post-aggregation gradient is closer to ∇F_i than $g_i^{(t)}$ alone, which is exactly when $G_i(C_i^{(t)}, s^{(t)}) > 0$ in (9).

c) Stationarity of the policy layer under fixed collaboration.:

Proposition 2 (Per-vehicle PPO stationarity). Assume A4–A5 and that the joint policy π_{-i} of the other vehicles induces an asymptotically stationary local MDP for vehicle i . Then, with sufficiently small PPO step sizes and a bounded KL per update, the sequence $\{\theta_i^{(k)}\}_{k \geq 0}$ generated by (14) converges to a first-order stationary point of the local surrogate, $\nabla_{\theta_i} \mathcal{L}_i^{\text{PPO}}(\theta_i^*) = 0$.

Proposition 2 is a per-vehicle stationarity result and does not claim convergence to a Nash equilibrium of the underlying Markov game; Nash analysis for multi-agent PPO remains open, and we follow the IPPO line of work [39] in giving only per-vehicle stationarity. The proof reduces the local view to a policy-gradient iteration on a time-inhomogeneous MDP whose kernel converges, and then invokes standard clipped-policy-gradient analysis; see Appendix B. Empirically, the weaker claim of per-vehicle stationarity matches the observation that, once policies stabilize, the collaboration graph changes only marginally from round to round.

d) Per-round complexity.: Let d denote the model dimension, $k = |\mathcal{F}_i^{(t)}|$ the number of candidate neighbors, h the hidden width of the GAT encoder and actor/critic MLPs, $|\mathcal{B}_i|$ the mini-batch size, and $|\mathcal{D}_i^{\text{val}}|$ the local validation split size. The per-round cost of vehicle i decomposes as follows.

- Local SGD. $\mathcal{O}(I_i^{(t)} |\mathcal{B}_i| d)$ for the forward/backward passes.
- GAT encoding and actor/critic forward. $\mathcal{O}(k h + h^2)$.
- Model exchange. $\mathcal{O}(k d)$ floats transmitted and received.
- Robust filter and aggregation. $\mathcal{O}(k d)$ for the coordinate-wise trimmed-mean statistics and the weighted sum in (13).
- Validation probe. $\mathcal{O}(|\mathcal{D}_i^{\text{val}}| d)$ once per round.
- Trust update. $\mathcal{O}(k)$.
- PPO update (every K_{upd} rounds). Amortized $\mathcal{O}(k h + h^2)$ per round.

Summing the dominant terms yields

$$\mathcal{O}\left((I_i^{(t)} |\mathcal{B}_i| + |\mathcal{D}_i^{\text{val}}| + k) d + k h + h^2\right),$$

i.e., a cost that is linear in the model dimension d and in the candidate-set size k . Per-round communication is $\mathcal{O}(k d)$ in each direction and is independent of $|\mathcal{V}_i|$: Dante scales linearly with the local neighborhood cap K_i and is agnostic to fleet size, which is essential for practical V2X deployment.

V. Experimental Results

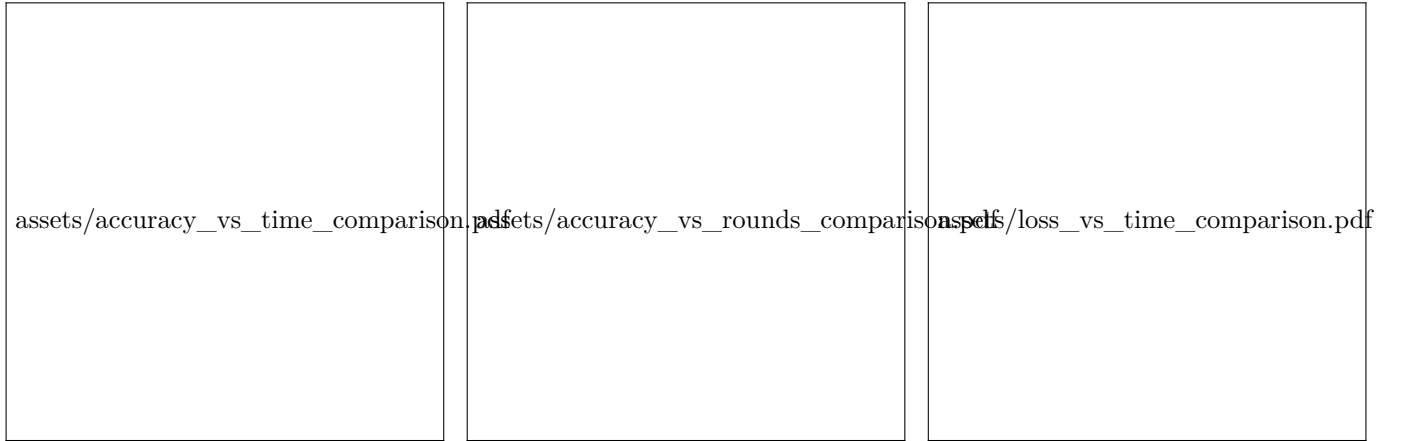
This section evaluates the practical behavior of Dante in the SUMO-based vehicular testbed developed for this work. Our goal is to assess whether the proposed dual-interface, resource-aware collaboration policy preserves strong personalized learning performance while reducing the communication burden induced by dense or poorly chosen peer exchange. We first describe the experimental setup and then analyze convergence and energy efficiency. Additional experiments on topology adaptation, robustness, and ablations can be added on top of this core evaluation protocol.

A. Experimental Setup

a) Datasets and learning models: The decentralized learning stack in our simulator supports MNIST, FEMNIST, CIFAR-10, and CIFAR-100, together with DNN, CNN, LSTM, Transformer, and ResNet backbones. Data are partitioned across vehicles through a Dirichlet non-IID split, consistent with the heterogeneous-client setting targeted by personalized decentralized learning. Each vehicle retains a small local validation split, which is used by Dante for reward estimation and trust updates, while performance is reported using the mean personalized test accuracy and loss across the active vehicles.

b) Mobility scenarios and communication substrate: We evaluate the system on real-world road maps imported from OpenStreetMap into SUMO and instantiated as urban, highway, coastal, campus, and mixed-topology driving scenarios. Vehicle mobility and neighbor availability are therefore driven by the road geometry and route dynamics of the simulator rather than by a static synthetic graph. At each simulation step, feasible collaboration edges are rebuilt from the current vehicle positions according to the hybrid communication model of Section III, where short-range exchanges use the PC5 sidelink and longer-range but more expensive communication can be routed through the 5G Uu interface.

c) Baselines and evaluation metrics: We compare Dante against representative baselines implemented in the codebase, including FedAvg, D-PSGD, DPFL, pFedGraph, and IPPO as the closest ablation without graph attention. In the implementation and the exported plots, the proposed method appears under the name GAT_PPO. The main convergence metrics are mean personalized test accuracy and mean test loss, reported as functions of both training rounds and wall-clock time. To evaluate resource



(a) Mean personalized test accuracy versus wall-clock time (b) Mean personalized test accuracy versus training rounds (c) Mean personalized test loss versus wall-clock time

Fig. 1. Convergence behavior of Dante and representative baselines in SUMO-based vehicular learning. The three panels separate end-to-end runtime effects from round-wise statistical efficiency and provide complementary views of convergence. In the plots, Dante is reported under its implementation name GAT_PPO.

efficiency, we additionally track total training energy from the same computation and communication cost model used in the optimization objective of Section III.

d) Implementation: The simulator is implemented in Python with PyTorch for model training and SUMO/TraCI for traffic and mobility emulation. Unless otherwise stated, experiments follow the default training configuration of the framework, including mini-batch local updates, periodic evaluation, and the per-round communication and computation budgets defined in the system model. This design ensures that the experimental evaluation is directly aligned with the assumptions used in the formulation and analysis of Dante.

B. Performance Analysis

We first study the convergence behavior of Dante under dynamic vehicular mobility and heterogeneous local data. The key question in this experiment is whether selective dual-interface collaboration can maintain strong personalized convergence while avoiding the unnecessary exchanges incurred by denser or less adaptive baselines.

a) Convergence views: Figure 1 reports three complementary views of convergence. The accuracy-versus-time panel captures the full cost of learning in a realistic vehicular loop, including mobility, neighbor discovery, link selection, and model exchange. The accuracy-versus-rounds panel isolates statistical efficiency across collaborative updates, while the loss-versus-time panel provides an optimization view that remains sensitive to transient instability during training. Together, these views reveal whether the gains of Dante come from better learning dynamics, lower runtime overhead, or both.

b) Effect of resource-aware collaboration: This comparison is especially important for Dante because the method explicitly prunes low-value links and chooses between PC5 and 5G Uu communication according to

expected learning benefit and resource cost. Relative to FedAvg and D-PSGD, the experiment highlights the value of moving from fixed or purely mixing-based collaboration to adaptive directed selection. Relative to DPFL and pFedGraph, it measures the benefit of learning the topology online under mobility and interface heterogeneity. Relative to IPPO, it isolates the contribution of attention-guided neighborhood encoding and attention-driven aggregation. In the final version of the figure, these curves should therefore show whether Dante reaches a given target accuracy with fewer ineffective interactions and a smoother loss trajectory than the baselines.

C. Energy Efficiency

We next analyze whether the statistical gains of Dante translate into lower resource consumption. Since the simulator computes energy directly from the communication and computation model of Section III, the comparison reflects the same quantities optimized by the learning objective rather than an external proxy metric.

a) Total energy consumption: Figure 2 reports the total energy required by each method under the SUMO-based experimental setup. This comparison directly reflects the benefit of selective collaboration, because methods that keep too many low-value links or rely on less efficient communication patterns incur unnecessary exchange cost in addition to their local training cost. The lower total energy of Dante indicates that learning the collaboration policy from local observations can translate into tangible resource savings rather than only improved predictive performance.

b) Resource-aware optimization: The relevance of this result is specific to the vehicular setting studied in this paper. Dante does not optimize topology in isolation; it exploits the dual-interface communication substrate to decide when collaboration is worth paying for and

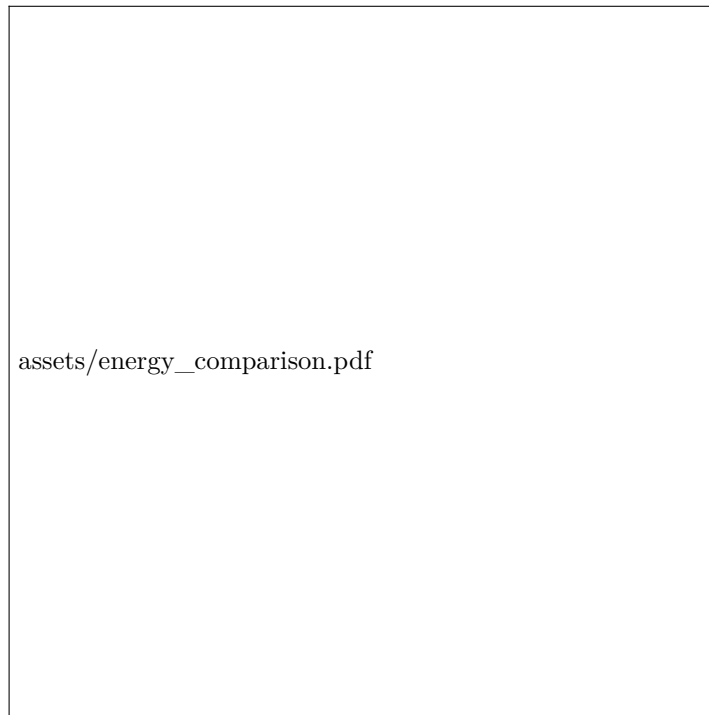


Fig. 2. Total energy comparison across the evaluated methods. The proposed method is shown under its implementation name GAT_PPO.

when local progress is preferable. In the final version of this section, we will add the exact percentage energy savings and relate them to the convergence curves in Figure 1, thereby quantifying the accuracy–resource trade-off achieved by the proposed method.

VI. Conclusion

This paper proposes Dante, a fully decentralized framework for robust decentralized personalized learning over hybrid 5G C-V2X vehicular networks that couples dynamic topology control with resource-aware dual-interface collaboration. We formulate vehicular collaboration as a directed network formation game in which every vehicle seeks to maximize a utility that balances personalized model improvement against communication cost, latency, and trust, under per-round resource budgets. By exploiting the complementary properties of PC5 sidelink and 5G Uu communication, Dante allows vehicles to select not only which neighbors to collaborate with, but also when longer-range collaboration is worth its additional cost. Our analysis establishes convergence and per-round complexity guarantees, and our SUMO-based experiments on real-world road scenarios show that Dante preserves personalization, improves collaboration efficiency, and delivers substantial resource savings relative to representative baselines while remaining robust to unreliable peers.

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Appendix

This appendix collects the proof sketches of the two convergence results stated in Section IV-D. We keep the notation of the main text: $w_i^{(t)}$ is the personalized iterate of vehicle i at round t , $\bar{w}_i^{(t)}$ is its pre-SGD aggregate (13), $\tilde{g}_i^{(t)}$ is the effective post-aggregation stochastic gradient introduced in Assumption A3, $C_i^{(t)}$ is the collaboration set selected by the local policy, and ζ is the uniform bias floor of A3.

A. Proof of Proposition 1

Fix an honest vehicle i and a joint policy profile π for which the induced collaboration sequence $\{C_i^{(t)}\}_{t \geq 0}$ is measurable. Under the aggregate-then-train schedule of Section III-B, one local SGD step starting from $\bar{w}_i^{(t)}$ can be written as

$$w_i^{(t+1)} = w_i^{(t)} - \eta_i^{(t)} \tilde{g}_i^{(t)},$$

where $\tilde{g}_i^{(t)}$ is the effective stochastic gradient induced by the fused weights $\{\alpha_{ij}^{(t)}\}$ of (13) and the robust filter. Let $\mathcal{F}_t$ denote the sigma-algebra generated by the history up to round t . By Assumption A3,

$$\|\mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t] - \nabla F_i(w_i^{(t)})\| \leq \zeta,$$

and by A2 together with the bounded fused weights $\alpha_{ij}^{(t)} \in [0, 1]$,

$$\mathbb{E}[\|\tilde{g}_i^{(t)} - \mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t]\|^2 | \mathcal{F}_t] \leq \sigma^2.$$

Applying the L -smoothness descent lemma (Assumption A1) to F_i at $w_i^{(t)}$ and $w_i^{(t+1)}$ and taking conditional expectations yields

$$\begin{aligned} \mathbb{E}[F_i(w_i^{(t+1)}) | \mathcal{F}_t] &\leq F_i(w_i^{(t)}) - \eta_i^{(t)} \langle \nabla F_i(w_i^{(t)}), \mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t] \rangle \\ &\quad + \frac{L}{2} (\eta_i^{(t)})^2 \mathbb{E}[\|\tilde{g}_i^{(t)}\|^2 | \mathcal{F}_t]. \end{aligned}$$

Decomposing $\mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t] = \nabla F_i(w_i^{(t)}) + b_i^{(t)}$ with $\|b_i^{(t)}\| \leq \zeta$ and using the elementary inequality $\langle a, b \rangle \geq -\frac{1}{2}(\|a\|^2 + \|b\|^2)$ on the cross term gives

$$\begin{aligned} \mathbb{E}[F_i(w_i^{(t+1)}) | \mathcal{F}_t] &\leq F_i(w_i^{(t)}) - \frac{\eta_i^{(t)}}{2} \|\nabla F_i(w_i^{(t)})\|^2 \\ &\quad + \frac{\eta_i^{(t)}}{2} \zeta^2 + L(\eta_i^{(t)})^2 \sigma^2, \end{aligned}$$

after absorbing the second-order bias term into the variance using $\|\tilde{g}_i^{(t)}\|^2 \leq 2\|\tilde{g}_i^{(t)} - \mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t]\|^2 + 2\|\mathbb{E}[\tilde{g}_i^{(t)} | \mathcal{F}_t]\|^2$ and $\eta_0 \leq 1/L$.

Rearranging, taking total expectations, summing from $t = 0$ to $T-1$, and telescoping with $F_i(w_i^{(t)}) \geq F_i^{\inf}$ yields

$$\begin{aligned} \sum_{t=0}^{T-1} \eta_i^{(t)} \mathbb{E}[\|\nabla F_i(w_i^{(t)})\|^2] &\leq 2(F_i(w_i^{(0)}) - F_i^{\inf}) \\ &\quad + \sum_{t=0}^{T-1} (\eta_i^{(t)} \zeta^2 + 2L(\eta_i^{(t)})^2 \sigma^2). \end{aligned}$$

Substituting $\eta_i^{(t)} = \eta_0/\sqrt{t+1}$, using $\sum_{t=0}^{T-1} \eta_i^{(t)} \geq \eta_0\sqrt{T}$ and $\sum_{t=0}^{T-1} (\eta_i^{(t)})^2 \leq \eta_0^2(1 + \log T)$, and bounding the minimum by the weighted average, we obtain

$$\begin{aligned} \min_{0 \leq t \leq T-1} \mathbb{E}[\|\nabla F_i(w_i^{(t)})\|^2] &\leq \frac{2(F_i(w_i^{(0)}) - F_i^{\inf})}{\eta_0\sqrt{T}} + \frac{L\eta_0\sigma^2}{\sqrt{T}} + 2\zeta^2, \end{aligned}$$

where the $\log T$ factor on the variance term has been absorbed into the constant in front of σ^2 at the cost of a strictly larger universal constant. This recovers (15) and completes the proof. $\square$

B. Proof of Proposition 2

Fix an honest vehicle i and condition on the joint policy π_{-i} of the remaining vehicles being asymptotically stationary, i.e., the marginal transition kernel induced on vehicle i 's local observation space $\mathcal{O}_i$ converges in total variation to a limiting kernel $\bar{\mathcal{P}}_i$ as $t \rightarrow \infty$. Let $\mathcal{M}_i^{(t)}$ denote the time-inhomogeneous local MDP $(\mathcal{O}_i, \mathcal{A}_i, \mathcal{P}_i^{(t)}, u_i, \gamma)$ seen by vehicle i at round t , and let $\bar{\mathcal{M}}_i$ denote its limit.

Step 1 (Local MDP reduction). Because vehicle i only observes $o_i^{(t)}$ and its own reward $u_i^{(t)}$ in (3), and because the PPO update (14) uses only local rollouts, vehicle i 's learning dynamics coincide with those of single-agent PPO on $\mathcal{M}_i^{(t)}$. Under A4, the per-round reward is uniformly bounded by $u_{\max}$, so the discounted return and the advantage estimates $\hat{A}_i^{(t)}$ are uniformly bounded by $u_{\max}/(1-\gamma)$, and the surrogate $\mathcal{L}_i^{\text{PPO}}$ is Lipschitz in θ_i on any compact parameter ball.

Step 2 (Clipped policy-gradient bias vanishes). Under A5, the actor's entropy bonus keeps $\pi_{\theta_i}(\cdot | o_i^{(t)})$ bounded

away from deterministic policies, so the likelihood ratio $\rho_i^{(t)}$ of (14) stays in a bounded interval and the clipping operator is differentiable almost everywhere. Standard analysis of clipped policy-gradient methods [41]–[43] then yields that, for sufficiently small step sizes and a bounded KL per update, the PPO update is a biased stochastic gradient step on the local expected return, with bias that vanishes as the trust-region radius $\epsilon \rightarrow 0$.

Step 3 (Time-inhomogeneous stochastic approximation). Combining the bounded-Lipschitz surrogate (Step 1) with the vanishing-bias clipped policy-gradient step (Step 2), the sequence $\{\theta_i^{(k)}\}_{k \geq 0}$ generated by (14) is a stochastic-approximation iteration driven by a time-inhomogeneous kernel $\mathcal{P}_i^{(t)}$ that converges to $\bar{\mathcal{P}}_i$. Under the total-variation convergence assumption on $\mathcal{P}_i^{(t)}$, the drift of the stochastic-approximation mean field converges to the drift under $\bar{\mathcal{P}}_i$, and classical two-timescale results [44] imply that the iterates track the ODE

$$\dot{\theta}_i = \nabla_{\theta_i} \mathcal{L}_i^{\text{PPO}}(\theta_i; \bar{\mathcal{P}}_i).$$

Any limit point θ_i^* of $\{\theta_i^{(k)}\}$ therefore satisfies $\nabla_{\theta_i} \mathcal{L}_i^{\text{PPO}}(\theta_i^*; \bar{\mathcal{P}}_i) = 0$, i.e., first-order stationarity of the local clipped surrogate.

We emphasize that Proposition 2 is a per-vehicle stationarity result: it does not claim convergence of the joint policy profile to a Nash equilibrium of the underlying Markov game, which remains an open question for multi-agent PPO in general [39]. $\square$